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Summary

MSc Data Science & AI candidate (École Polytechnique & HEC Paris, GPA 3.8/4.0), Stanford visiting researcher applying ML to large-scale time-series prediction, and Quantitative Researcher Intern at **WorldQuant**.

Education

École Polytechnique — MSc Data Science & AI (Year 1) GPA: 3.8/4.0 *Sep 2024 – Jun 2025*
– Coursework: Probability & Statistics, Linear Algebra, Optimization, Deep Learning, Machine Learning.

HEC Paris — MSc Data Science & AI (Year 2) GPA: 3.85/4.0 *Sep 2025 – Jun 2026*
– Coursework: ML for Financial Markets, Time Series, Causal Inference, Reinforcement Learning.

INP–ENSEEIHT, Toulouse — French engineering school, Engineer's degree in Computer Science *2020 – 2023*
– Coursework: Probability, Statistics, Optimization, Algorithms, High-Performance Computing.

CPGE PCSI/PSI, Lycée Alphonse Daudet — Intensive prep classes: Maths, Physics, Engineering, Chemistry *2017 – 2020*

Experience

Quantitative Researcher Intern | *WorldQuant (Millennium affiliate), Paris* *Oct 2026 – Mar 2027*
– 6-month internship in a Millennium systematic pod; research on **LLM-based signal extraction** from text data.

Visiting Student Researcher | *Stanford University, Stanford, CA* *Mar 2026 – Sep 2026*
– First-author study on medical time-series data: predicted a rare post-surgery cardiac complication from the records and bedside signals of 7,653 patients, reaching **AUROC 0.74**.
– Tested out of sample and **externally validated on unseen data** from a second hospital system (AUROC 0.695; 0.731 after recalibration).

Data Scientist Intern | *Crédit Agricole CIB (CACIB), Paris* *Mar 2025 – Sep 2025*
– Predicted late implementation of audit recommendations across 200k+ records (logistic regression + XGBoost): **F1** $\approx$ **0.75**, validated out-of-time with domain experts.

Quantitative Research & Engineering Projects

Crypto Prediction-Market Strategy | *Personal Project, live* *2026 – present*
– Systematic strategy on Polymarket crypto markets, **trading live** with an automated daily data and signal pipeline.

Crypto Perpetuals Alpha Strategy | *ML for Financial Markets, HEC Paris* *2026*
– Ranked **#1 of 13 teams** on out-of-sample Sharpe: dollar-neutral long-short across 411 crypto perpetual futures (gross of costs).
– 379 features pruned to 316 (Spearman IC, correlation), Ridge + XGBoost under a **rolling 12-month walk-forward**; models refit per window, IC filter frozen on train as an explicit anti-leakage choice.

Cognitive Alpha — Spatial Decision Analytics, World Cup 2022 | *Personal Project* *2025–2026*
– Per-pass gap between the player's choice and the spatial optimum: xT trained by Markov-chain value iteration on **126k possession actions**, NumPy pitch-control engine, Streamlit dashboard over ∼5.4 GB of tracking data.

Competitions

– **QRT / ENS Data Challenge 2026** (ongoing) — **ranked 2nd of 1,200+ participants** on next-day return signs of anonymised allocations; 0.51 benchmark to 0.79 accuracy by reconstructing their holdings from external data.
– **G-Research Trading Game, Paris (Sep 2026)** — **1st place** in one of the rounds vs ∼30 teams: analytic optimal-stopping threshold, adapted live to an unknown reward distribution and competing agents.

Technical Skills & Activities

Programming: Python (primary), SQL (PostgreSQL), R, MATLAB, Bash; C++ / Java (working knowledge)
Quantitative & ML: Probability & Statistics, Time-Series, Survival Analysis, Signal/Alpha Research
Tools: PyTorch, XGBoost, Scikit-learn, NumPy, Pandas, MLflow, Docker, Git, Kafka, FastAPI, Linux
Open source: contributor to mplsoccer (`Pitch.pcolormesh`, merged); PR open on statsbombpy
Self-study: Stochastic calculus & options theory (Hull, Natenberg, Wilmott); Akuna Options 101 & 202; Jane Street puzzles
Languages: French (native), English (fluent); Dutch, Spanish, Portuguese & Arabic (B2)
Activities: Built and sold a 200k-follower X/Twitter account to Polymarket; maths tutor (3+ years); chess (1700 Elo)